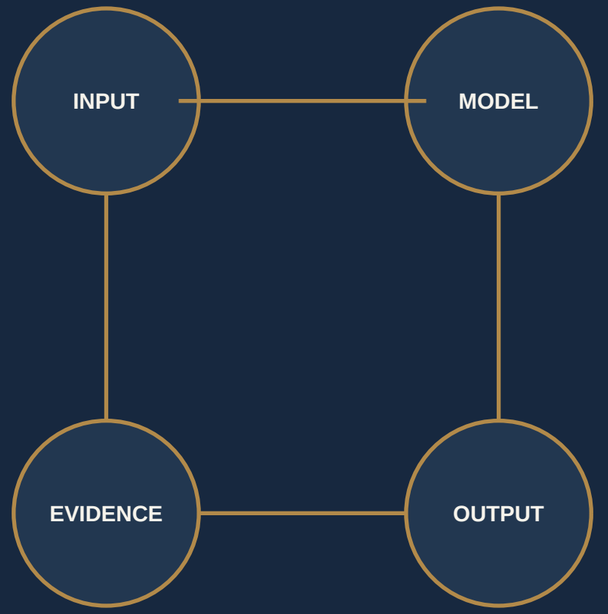
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Control Systems

AI.106 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define feedback.
- Apply key methods including open-loop and closed-loop systems.
- Build or analyze a practical system involving transfer functions.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Feedback

02

METHODS

Open-loop and closed-loop systems

03

SYSTEMS

Transfer functions

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Feedback

Feedback

02 · Open-loop and closed-loop sys

Open-loop and closed-loop systems

03 · Transfer functions

Transfer functions

04 · Dynamic systems

Dynamic systems

05 · Stability

Stability

06 · PID control

PID control

07 · Sensors

Sensors

08 · Actuators

Actuators

Core ideas, made explicit

AI.106 • Distinguish definitions, assumptions and evidence

- Feedback
- Open-loop and closed-loop systems
- Transfer functions
- Dynamic systems
- Stability

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.106 • Distinguish definitions, assumptions and evidence

- PID control
- Sensors
- Actuators
- State-space representations and digital control
- Block diagrams

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Feedback

INPUTS
Open-loop and closed-loop systems

METHOD
Transfer functions

OUTPUT
Dynamic systems

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving feedback.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using feedback.
- Implement or simulate a baseline using open-loop and closed-loop systems.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.